

First Order Logic (FOL)

First Order Logic is a method used in AI to represent knowledge about objects, their properties, and relationships between objects.

It is more powerful than propositional logic because it can express general statements.

Syntax of First Order Logic

1. Constants: Represent specific objects (Ram, Sita).
2. Variables: Represent general objects (x, y, z).
3. Predicates: Represent properties or relations like Human(x), Loves(x, y).
4. Functions: Return objects like Father(x).
5. Quantifiers: $\forall$ (for all), $\exists$ (there exists).

Semantics of First Order Logic

Semantics defines the meaning and truth of statements.

It includes domain (set of objects) and interpretation (truth of statements).

Example

1. $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

2. $\text{Human}(\text{Ram})$

Conclusion: $\text{Mortal}(\text{Ram})$